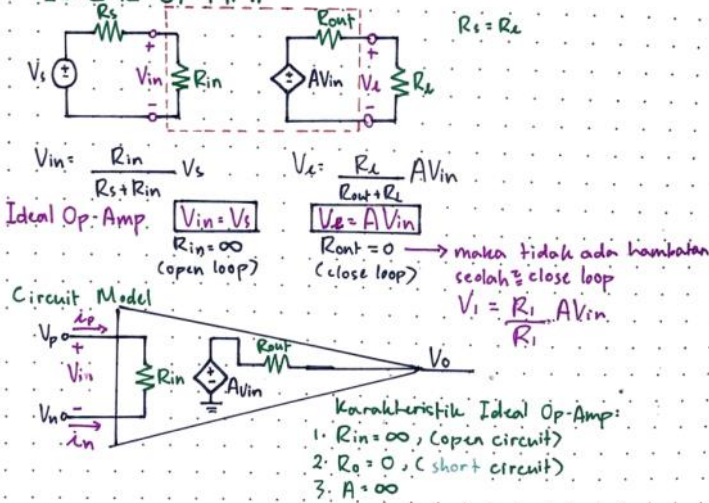


Soal UAS DRL 2021

07 January 2023 10:02

① IDEAL OP-AMP



Karakteristik Op-Amp:

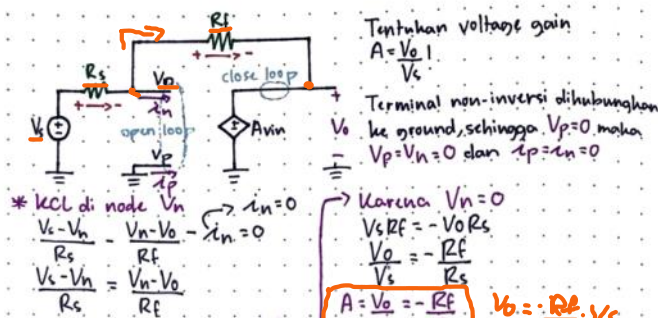
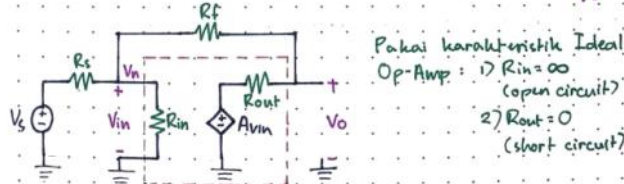
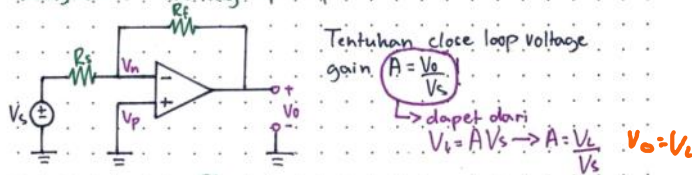
- 1) Tegangan yang melewati V_p dan V_n adalah nol
 $V_{in} = V_p - V_n$
 $V_p = V_n$
 * Tegangan terjadi jika ada perbedaan potensial
 $V_{in} \rightarrow$ tidak ada perbedaan potensial
- 2) Karena $R_{in} = \infty$ maka arus yang masuk V_p dan V_n dianggap nol
 $I_p = \frac{V_p}{R_{in}} = 0$
 $I_n = \frac{V_n}{R_{in}} = 0$
 $I_p = I_n = 0$
 * Tegangan masih ada tapi arus = 0 karena $R = \infty$
 arus ga bisa lewat perumpamaannya: $V_n =$ pipa, Arus = air mengalir

② INVERTING AMPLIFIER

Sinyalnya dibalik (+ - nya) ✓ mempermudah skematika



Rangkaian Inverting Op-Amp



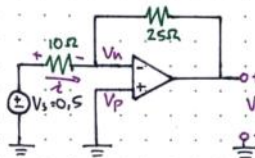
$$\frac{V_s - V_n}{R_s} - \frac{V_n - V_o}{R_f} - i_n = 0$$

$$\frac{V_s - V_n}{R_s} = \frac{V_n - V_o}{R_f}$$

$$V_s R_f - V_n R_f = V_n R_s - V_o R_s$$

$$V_s R_f = V_n (R_s + R_f) - V_o R_s$$

$$\frac{V_o}{V_s} = -\frac{R_f}{R_s} \quad V_o = -\frac{R_f}{R_s} V_s$$



1. Tentukan V_o
2. Tentukan arus i
3. Berapa kali penguatannya (A)

$$1) \frac{V_o}{V_s} = -\frac{R_f}{R_s}$$

$$V_o = -\frac{R_f}{R_s} V_s$$

$$= -\frac{25}{10} \cdot \frac{1}{2}$$

$$V_o = -1,25V$$

$$2) i = \frac{V_s - V_n}{R_s}$$

$$= \frac{0,5 - 0}{10}$$

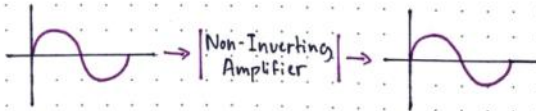
$$i = 0,05A$$

$$3) A = \frac{V_o}{V_s}$$

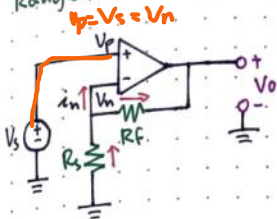
$$= \frac{-1,25}{0,5}$$

$$A = -2,5$$

③ NON-INVERTING AMPLIFIER



Rangkaian Non-Inverting Op-Amp



Tentukan voltage gain $A = \frac{V_o}{V_s}$

Terminal non-inversi dihubungkan ke sumber, sehingga $V_p = V_s$ maka $V_p = V_n = V_s$

* KCL di node V_n

$$0 - \frac{V_n}{R_s} - \frac{V_n - V_o}{R_f} - i_n = 0$$

Op-Amp Ideal $i_p = i_n = 0$

$$-\frac{V_n}{R_s} = \frac{V_n - V_o}{R_f}$$

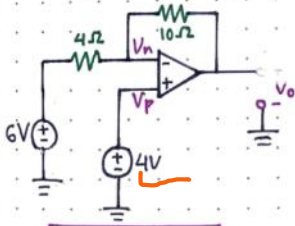
dan $V_p = V_n = V_s$

$$-\frac{V_s}{R_s} = \frac{V_s - V_o}{R_f}$$

$$-V_s R_f = V_s R_s - V_o R_s$$

$$V_s (R_f + R_s) = V_o R_s$$

$$A = \frac{V_o}{V_s} = \frac{R_f + R_s}{R_s}$$



$$A = \frac{V_o}{V_s} = \frac{-1}{6} = -\frac{1}{6}$$

Tentukan V_o , A, dan arus di 4Ω !

$$V_p = V_n = 4V$$

$$V_s = 6V$$

* KCL di node V_n

$$\frac{6 - V_n}{4} = \frac{V_n - V_o}{10}$$

$$\frac{6 - 4}{4} = \frac{4 - V_o}{10}$$

$$\frac{1}{2} = \frac{4 - V_o}{10}$$

$$5 = 4 - V_o$$

$$V_o = -1V$$

* Arus di 4Ω

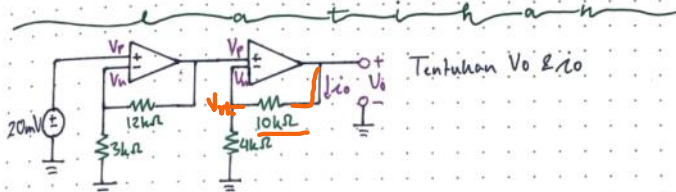
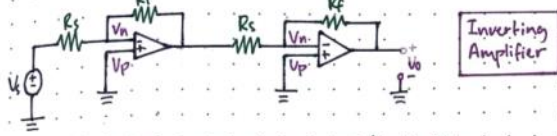
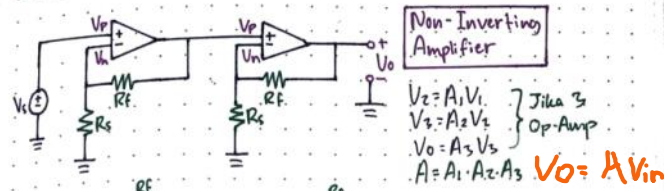
$$i = \frac{V_s - V_n}{4}$$

$$= \frac{6 - 4}{4}$$

$$= \frac{2}{4}$$

$$= 0,5A$$

① CASCADED AMPLIFIER



* Op-Amp Pertama

$$V_o = \frac{3k + 12k}{3k} (20m)$$

$$= 100mV$$

* Op-Amp Kedua

$$V_o = \frac{10k + 4k}{4k} (100m)$$

$$= 350mV$$

inverting $V_o = \frac{R_f}{R_s} V_i$ | Non-inverting $V_o = \frac{R_f + R_s}{R_s} V_i$

* i_o yang dibutuhkan melewati resistor 10kΩ

$$i_o = \frac{V_o - V_{in}}{10k}$$

$V_{in} = V_p = 100mV$, maka:

$$i_o = \frac{350m - 100m}{10k} = \frac{(350 - 100) \times 10^{-3}}{10 \cdot 10^3} = 25\mu A$$

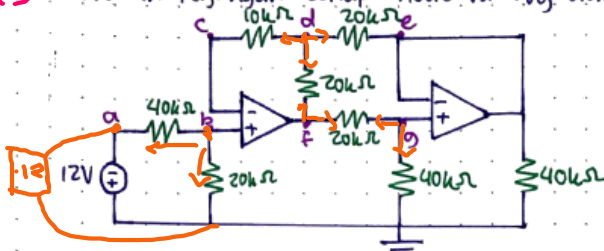
UAS DRL

21 Januari 2022

Nama: Alvina Patricia Wijayan
NPM: 6152101003

UAS 1. Tentukan tegangan setiap node $V_a \dots V_g$ dengan hitungan secara lengkap!

Analisis node



1* Tinjau node V_a

$V_a = -12V$ karena supernode thd ground

2* Tinjau node V_b

$$\frac{V_b - V_a}{40 \cdot 10^3} + \frac{V_b}{20 \cdot 10^3} = 0 \rightarrow \frac{V_b - V_a + 2V_b}{40 \cdot 10^3} = 0$$

$$3V_b = V_a$$

$$3V_b = -12$$

$$V_b = -12 : 3 = -4V$$

3* Tinjau node V_c

$V_p = V_n \rightarrow V_b = V_c$ karena op-amp ideal, maka $V_c = -4V$

4* Tinjau node V_d

$$2 \frac{V_d - V_c}{10 \cdot 10^3} + \frac{V_d - V_e}{20 \cdot 10^3} + \frac{V_d - V_f}{20 \cdot 10^3} = 0$$

$$2V_d - 2V_c + V_d - V_e + V_d - V_f = 0 ; V_c = -4V$$

$$4V_d + 8 - V_e - V_f = 0$$

$$-4V_d + V_e + V_f = 8 \dots (1)$$

5* Tinjau node V_f

$$\frac{V_f - V_g}{20 \cdot 10^3} + \frac{V_f - V_d}{20 \cdot 10^3} = 0 ; V_e = V_g \text{ karena op-amp ideal}$$

$$2V_f - V_e - V_d = 0 \dots (2)$$

6* Eliminasi pers (1) dan (2)

$$-4V_d + V_e + V_f = 8$$

$$-V_d - V_e + 2V_f = 0 +$$

$$-5V_d + 3V_f = 8 \dots (6)$$

7* Tinjau node V_g

$$\frac{V_g - V_f}{20 \cdot 10^3} + \frac{V_g}{40 \cdot 10^3} = 0$$

$$2V_g - 2V_f + V_g = 0$$

$$3V_g = 2V_f$$

$$V_f = \frac{3}{2} V_g ; V_g = V_e$$

$$V_f = \frac{3}{2} V_e \dots (3)$$

8* Substitusi (3) ke (1) dan (2)

$$(1) -4V_d + V_e + \frac{3}{2} V_e = 8 \quad (2) 2 \cdot \frac{3}{2} V_e - V_e - V_d = 0$$

$$-4V_d + \frac{5}{2} V_e = 8 \dots (4)$$

$$2V_e - V_d = 0 \dots (5)$$

10* $V_e = V_g$, maka $V_g = 1,45V$

$$(3) V_f = \frac{3}{2} V_e = \frac{3}{2} \cdot 1,45 = 2,175V$$

9* Eliminasi (4) dan (5)

$$-4V_d + \frac{5}{2} V_e = 8 \quad \times 1 \quad -4V_d + \frac{5}{2} V_e = 8$$

$$-V_d + 2V_e = 0 \quad \times 4 \quad -4V_d + 8V_e = 0 -$$

$$5,5V_e = 8$$

$$V_e = 1,45V$$

11* Substitusi nilai V_f ke pers (6)

$$-5V_d + 3V_f = 8$$

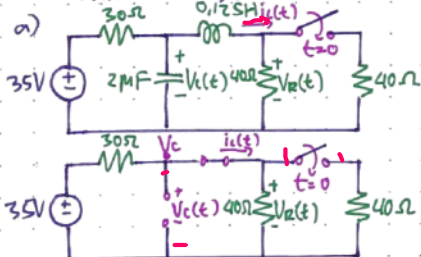
$$-5V_d + 3(2,175) = 8$$

$$6,525 - 8 = 5V_d$$

$$-1,475 = 5V_d \rightarrow V_d = -0,295V$$

Uts

2. Tentukan V_R , V_C dan i_L sebelum dan setelah saklar dibuka C open L short



1* $t < 0$

$$i = \frac{V}{R_{eq}} = \frac{35}{30+40} = 0,5A$$

$$i_L = i = 0,5A$$

$$V_R = i \cdot R = 0,5 \cdot 40 = 20V$$

$$V_C = 35 - (30)(0,5) = 35 - 15 = 20V$$

2* $t > 0$

$$R_{eq} = (40 \parallel 40) + 30 = \frac{40 \cdot 40}{40+40} + 30 = 20 + 30 = 50\Omega$$

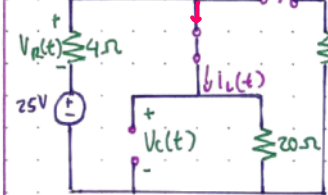
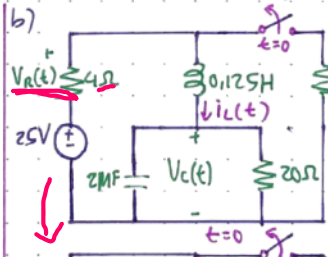
$$i = \frac{V}{R_{eq}} = \frac{35}{50} = 0,7A$$

$$V_C = 35 - (30)(0,7) = 35 - 21 = 14V$$

$i_L = i = 0,7A$

$$i_R = \frac{40}{40+40} \cdot 0,7 = 0,35A$$

$$V_R = i \cdot R = 0,35 \cdot 40 = 14V$$



1* $t < 0$

$$i = \frac{V}{R_{eq}} = \frac{25}{20} = 1,25A$$

$$V_R = 4 \cdot 1,25 = 5V$$

$$i_L = \frac{80}{20+80} \cdot 1,25 = 1A$$

$$R_{eq} = \left(\frac{20 \cdot 80}{20+80} \right) + 4 = \frac{1600}{100} + 4 = 16 + 4 = 20\Omega$$

$$V_C = V_{20\Omega} \text{ karena paralel} = 1 \cdot 20 = 20V$$

2* $t > 0$

$$R_{eq} = 4 + 20 = 24$$

$$i = \frac{V}{R} = \frac{25}{24} = 1,04A$$

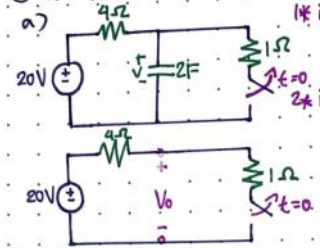
$$i_L = i = 1,04A$$

$$V_R = i \cdot R = (1,04)(4) = 4,16V$$

$$V_C = V_{20\Omega} \text{ karena paralel} = i \cdot R = (1,04)(20) = 20,8V$$

Pertemuan 13 Responsi - 22 Desember 2021

1) Tentukan V_C untuk $t < 0$ dan $t > 0$ w/ setiap sirkuit



3* $T = R_{eq} \cdot C \rightarrow R_{eq} = 4$ karena $= (4)(2) \Omega$ berada di $=$ setelah rangkaian open

4* Maka

$$V_C(t) = 20 - (20 - 4)e^{-\frac{1}{4}t}$$

$$= 20 - 16e^{-0,25t}$$

1* i) $t < 0$ close

$$V_C(0) = 20 \times \left(\frac{1}{1+4} \right) = 4V$$

2* ii) $t > 0$ open

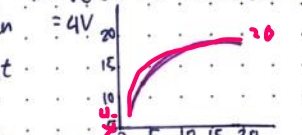
$$V_C(t) = V_s + (V_0 - V_s)e^{-t/\tau}$$

$$= A - (A - B)e^{-t/\tau}$$

dimana:

$$A = V_C(t \rightarrow \infty) \text{ (Rangkaian open)} = 20V$$

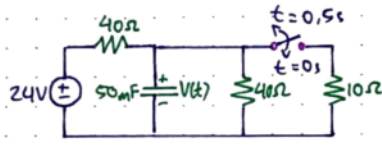
$$B = V_C(t \rightarrow 0^+) = V_C(t \rightarrow 0^-) = 4V$$



3_Alvina

UAS

3. Rangkaian telah steady state sblm saklar terbuka pada $t=0$. Saklar tetap terbuka selama 0,5 detik kemudian saklar kembali tertutup.
 • Persamaan $V(t)$ untuk $t \geq 0$
 • Gambarkan grafik



1* $t < 0$

$$R_{eq} = (40 \parallel 10) + 40$$

$$= \frac{40 \cdot 10}{40 + 10} + 40$$

$$= 8 + 40 = 48 \Omega$$

$$i = \frac{V}{R_{eq}} = \frac{24}{48} = 0,5 A$$

$$V(0^-) = 24 - (0,5)(40)$$

$$= 24 - 20$$

$$= 4V$$

$$V(0^+) = V(0^-) = V(0) = 4V$$

2* $0 < t < 0,5$

$$R_{eq} = 40 + 40 = 80 \Omega$$

$$i = \frac{V}{R_{eq}} = \frac{24}{80} = 0,3 A$$

$$V(t) = A - (A - B)e^{-t/\tau}$$

$A = V(t \rightarrow +\infty)$ jika switch terbuka
 $= 40 \cdot i \rightarrow$ karena $V_c = V_{40\Omega}$ karena paralel

$$= 40 \cdot 0,3$$

$$= 12V$$

$B = V(t \rightarrow 0) = V(0^+) = 4V$

$$\tau = R \cdot C \rightarrow R_{eq} = \frac{40 \cdot 40}{40 + 40}$$

$$= 20 \cdot 50 \cdot 10^{-3}$$

$$= 1 \text{ detik}$$

$$V(t) = 12 - (12 - 4)e^{-t/1}$$

$$= 12 - 8e^{-t} V$$

3* $0,5 < t < +\infty \rightarrow$ sama spt gambar 1

$$V(0,5) = 12 - 8e^{-0,5}$$

$$= 12 - 4,85$$

$$= 7,15V$$

$$V(t) = A - (A - B)e^{-t/\tau}$$

$A = V(t \rightarrow +\infty)$
 $= 4V \rightarrow$ karena kembali ke V semula

$B = V(t \rightarrow 0,5)$
 $= 7,15V$

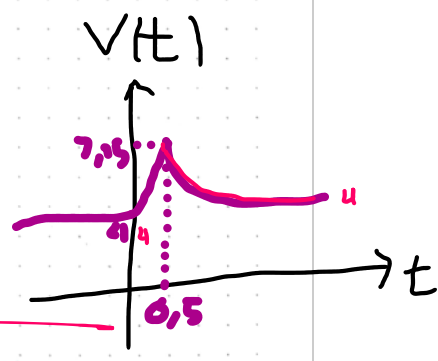
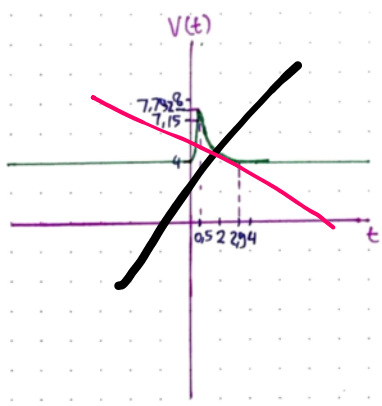
$$\tau = R \cdot C \rightarrow R_{eq} = (40 \parallel 10) + 40$$

$$= 48 \cdot 50 \cdot 10^{-3}$$

$$= 2,4 \text{ detik}$$

$$V(t) = 4 - (4 - 7,15)e^{-t/2,4}$$

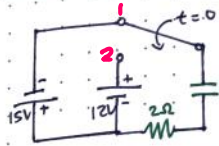
$$= 4 + 3,15e^{-0,416t} V$$



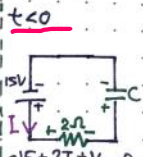
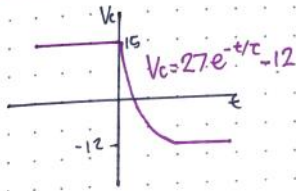
$\tau = 0,632 V$
 $= 0,632(12)$
 $= 7,584 \rightarrow 1 \text{ detik}$
 $\downarrow 0,5 \text{ detik}$
 $3,792V + 4 = 7,792V$

$$V(t) = \begin{cases} 4V & t < 0 \\ 12 - 8e^{-t} V & 0 < t < 0,5 \\ 4 + 3,15e^{-0,416t} V & 0,5 < t < +\infty \end{cases}$$

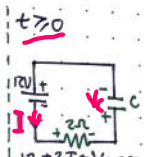
kontor 3



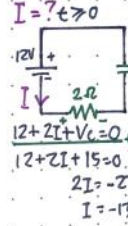
Pada saat $t < 0 \rightarrow V_c = 15V$
 Pada saat $t = 0 \rightarrow V_c = 15V$
 Pada saat $t > 0 \rightarrow V_c = -12V$



$$\begin{aligned} -15 + 2I + V_c &= 0 \quad t < 0 \\ -15 + 2 \cdot 0 + V_c &= 0 \\ V_c &= 15V \end{aligned}$$



$$\begin{aligned} 12 + 2I + V_c &= 0 \quad t > 0 \\ 12 + 2 \cdot 0 + V_c &= 0 \\ V_c &= -12V \end{aligned}$$



$$\begin{aligned} 12 + 2I + V_c &= 0 \quad t = \infty \\ 12 + 2I + 15 &= 0 \\ 2I &= -27A \\ I &= -13,5A \end{aligned}$$

$$\begin{aligned} t < 0 &\rightarrow I = 0A \\ t = 0 &\rightarrow I = -13,5A \\ t > 0 &\rightarrow I = 0A \end{aligned}$$

